Saurabh Belgaonkar

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EDUCATION

TEXAS A&M UNIVERSITY 2023 - 2027
Ph.D. Mechanical Engineering GPA: 4.0/4.0
Advisor: Dr.Sivakumar Rathinam

IISC BANGALORE Oct 2020 – Jul 2022
M.Tech Mechanical Engineering
GPA: 9.2/10

NIT WARANGAL Aug 2015 – May 2019
B.Tech Mechanical Engineering
GPA: 8.92/10

SKILLS

Programming & Systems
C++, Python, ROS2, Linux, Git, SQL, OpenCV, OMPL, Databricks, Spark

Perception & ML
PyTorch, CUDA, TensorRT, OpenVINO, Stable-Baselines3, Gymnasium, MuJoCO

Robotics & Sensing
Camera–LiDAR calibration, sensor fusion, multi-object tracking, segmentation

Optimization
Gurobi, CPLEX, MIP, convex optimization, dynamic programming, LKH

SELECTED COURSES

Convex Optimization, Reinforcement Learning, Intelligent Systems, Control Systems, Multi-modal Computer Vision, Algorithms

PUBLICATIONS

- J. Keshavan, **S. Belgaonkar**, "Adaptive Super-Twisting Control of a First-Order Sliding Mode with an Output Constraint," *IEEE Access*.
- **S. Belgaonkar** et al., "A Mixed-Integer Convex Optimization based approach for Hybrid UAV Path Planning in Noise-Restricted Zones," *IEEE Transactions on Aerospace and Electronic Systems*.
- Under review: **S. Belgaonkar** et al., "Pareto Front Approximation for the Biobjective Steiner Traveling Salesman Problem," *IROS*.

TECHNICAL STRENGTHS

- Real-time **C++/ROS2** autonomous driving software
- Camera–LiDAR perception, sensor fusion, and **SLAM** pipelines
- **MIP** and convex optimization with **Gurobi**
- ML model training, edge deployment, and inference acceleration

ACHIEVEMENTS

- Led an autonomous car competition team of 35 to consecutive 2nd Place finishes Autodrive Challenge II(2024,2025).
- Awarded the **S. V. Sastry Memorial Gold Medal**, IISc Bangalore, 2022.

PROFESSIONAL EXPERIENCE

CAPTAIN; PERCEPTION LEAD Oct 2023 – Present
GM SAE AutoDrive Challenge
- Enabled reliable object detection and tracking under **adverse conditions** (fog) using a **C++/ROS2** multi-sensor perception stack with **LiDAR, radar, and cameras**.
- Optimized edge deployment of **RF-DETR**-based object detection, **U-Net**-based lane segmentation, **PointPillars**-based LiDAR detection, and LiDAR clustering, increasing perception throughput by approximately **50%** to **15 Hz**.
- Implemented object-aware navigation with **Nav2 Behavior Trees**, using perception outputs to define **modular** obstacle-avoidance and scene-specific behaviors.
- Enabled autonomous operation in **GPS-denied** environments by integrating **IMU, GNSS, cameras, and LiDAR** for pose estimation with **LiDAR SLAM**.

GRADUATE RESEARCH ASSISTANT Aug 2023 – Present
Autonomous Systems Lab, Texas A&M University
- Developed **optimization** algorithms for hybrid UAV planning and TSP routing under battery, safety, and **environmental constraints**.
- Applied problem-specific solvers across planning tasks, including **A***, **branch-and-cut MILP**, and **LKH** for routing, and **SCQP** for multi-robot trajectory planning.

DATA SCIENTIST Aug 2022 – Jul 2023
Atlas Copco, India
- Built machine-health monitoring pipelines with **CNNs**, **GANs**, signal processing, and **Databricks** enabling early fault prediction that reduced unexpected downtime.
- Built a vibration-based compressor speed estimator using 1D CNN features, and **LightGBM**, improving accuracy from **70%** to **99.5%** using limited labeled data.

SELECTED PROJECTS

REAL-TIME PERCEPTION AND SENSOR FUSION Feb 2024 – Present
GM SAE AutoDrive Challenge
- Deployed transformer-based **RF-DETR** object detector in **TensorRT** across a **multi-camera C++/ROS2** pipeline, reducing deployment runtime by around **60%**.
- Deployed a **PointPillars**-based LiDAR detector for pedestrians, vehicles, barrels, and animals, with **Kalman filtering** for stable **multi-object tracking**.
- Fused camera and LiDAR detections using projected-box **IoU** and **Hungarian** assignment, improving adverse-condition detection by **32%**.
- Built a **YOLOv11**-based Siamese RGB–LiDAR detector with feature-level fusion, improving detection by **45%** over a camera-only baseline under adverse conditions.

OPTIMIZATION-BASED ENERGY-AWARE ROUTING Feb 2024 – Jan 2025
Autonomous Systems Lab, Texas A&M University
- Formulated continuous-space path and energy planning for **hybrid UAVs** under **battery** and electric-only **noise-restricted-zone** constraints and solved it as **mixed-integer convex program (MICP)**, achieving up to **10**× speedup over the baseline.
- Extended the MICP framework to **TSP**, discretizing state-of-charge to convert the problem into **GTSP** and solving with **LKH** for **5**× improvement over baseline.
- Developed a load-dependent energy model and **branch-and-cut MILP** solver for **Load dependent TSP**, solving 50-target instances to optimality in under 1 minute.

RISK-AWARE BIOBJECTIVE TSP IN THREAT ZONES Mar 2025 – Mar 2026
Autonomous Systems Lab, Texas A&M University
- Formulated threat zones TSP planning as a **Biobjective Steiner TSP** and developed **two-phase pareto local search** to approximate distance–risk Pareto fronts.
- Initialized the search with **LKH** anchors, then applied **2-opt/3-opt** multi-expansion and tolerance-guided heuristics, reducing runtime by **90%** vs. the exact baseline.

SCALABLE MULTI-ROBOT TRAJECTORY PLANNING Aug 2025 – Nov 2025
Autonomous Systems Lab, Texas A&M University
- Developed a **DCA**-based planner for **multi-robot** trajectory planning with obstacle and inter-robot collision avoidance, solved via sequential convex **QP** subproblems.
- Scaled collision-free planning with static obstacles to **160 robots**, reducing computation time by **28%** over the SOTA baseline.